

-Total marks – 120

Attempt All Questions

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra booklets are available.

Question 1 (15 Marks)

Marks

(a) Find $\int x^2 \sin(x^3) dx$. 2

(b) Use integration by parts to evaluate $\int_0^1 \tan^{-1} x dx$. 3

(c) (i) Find the real numbers a and b such that $\frac{x}{(x-1)(x+4)} \equiv \frac{a}{x-1} + \frac{b}{x+4}$. 2

(ii) Find $\int \frac{x}{(x-1)(x+4)} dx$. 2

(d) Find $\int \frac{x+4}{x^2-4x+13} dx$. 3

(e) Use the substitution $t = \tan \frac{x}{2}$ to evaluate

$\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \cos x}$ 3

Question 2 (15 Marks) Use a SEPARATE writing booklet.**Marks**

(a) Let $w = 1 + i$ and $z = 1 - i\sqrt{3}$, simplify the following

(i) $\overline{wz}$ **1**

(ii) $\frac{1}{w}$ **1**

(iii) $\frac{i(\operatorname{Re}(z) - z)}{\operatorname{Im}(z)}$. **2**

(b) Sketch the region on the Argand diagram where the inequalities $|z| \leq 2$ and

$\pi \geq \arg z \geq -\frac{\pi}{4}$ hold simultaneously. **3**

(c) Solve the equation $x^2 - 4x + (1 - 4i) = 0$. Answer should be expressed in the form $a + ib$ **4**

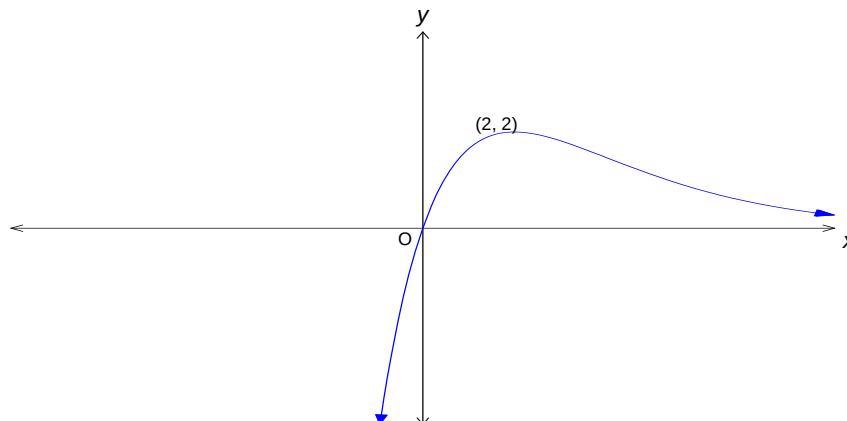
(d) The complex number $z = x + iy$, where x and y are real, satisfies the parametric equation $z = 1 + 2i + t(3 - 4i)$ where t is a real parameter.

(i) Show that the Cartesian equation of the locus of the point P which represents z in an Argand diagram is given by $4x + 3y = 10$. **2**

(ii) Hence find the minimum value of $|z|$. **2**

Question 3 (15 Marks) Use a SEPARATE writing booklet.**Marks**

- (a) The curve shown in the diagram is the equation $y = f(x)$. There is a maximum turning point at $(2, 2)$ and the curve crosses the x axis at $(0, 0)$. The graph has a horizontal asymptote at $y = 0$.

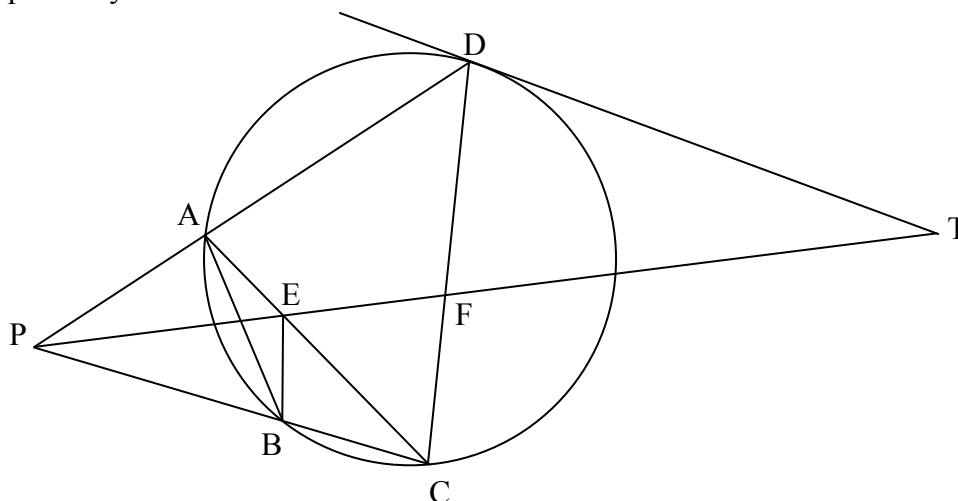


Sketch the following curves on separate diagrams, showing all of the essential features.

- | | | |
|-------|---|----------|
| (i) | $y = f(x+2)$ | 1 |
| (ii) | $y = \frac{1}{f(x)}$ | 2 |
| (iii) | $y = (f(x))^2$ | 2 |
| (iv) | $y = -x \times f(x)$ | 2 |
| (b) | (i) Show that $P(x) = x^5 - 3x^4 + 4x^3 - 4x^2 + 3x - 1$ has $x = 1$ as a root of multiplicity 3. | 2 |
| | (ii) Verify that $x = i$ is also a root of $P(x)$. | 1 |
| | (iii) Hence solve the equation $P(x) = 0$. | 2 |
| (c) | Let α, β, γ be the roots of the equation $x^3 - 2x^2 - 5x - 1 = 0$. Form an equation whose roots are $\frac{1}{\sqrt{\alpha}}, \frac{1}{\sqrt{\beta}}$ and $\frac{1}{\sqrt{\gamma}}$. | 3 |

Question 4 (15 Marks) Use a **SEPARATE** writing booklet.**Marks**

- (a) For what values of k does the equation $\frac{x^2}{5-k} + \frac{y^2}{k-3} = 1$ represent:
- (i) a circle? 2
- (ii) a hyperbola? 2
- (b) The points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ are points on the rectangular hyperbola $xy = c^2$. Tangents to the rectangular hyperbola at P and Q intersect at the point $R(X, Y)$.
- (i) Show that the tangent to the rectangular hyperbola at any point $T\left(ct, \frac{c}{t}\right)$ has equation $x + t^2y - 2ct = 0$. 1
- (ii) Find the coordinates R . 2
- (iii) If P and Q are variable points on the rectangular hyperbola which move so that $p^2 + q^2 = 2$, show that the equation of the locus of R is given by $xy + y^2 = 2c^2$. 3
- (c) $ABCD$ is a cyclic quadrilateral. DA is produced and CB produced meet at P . T is a point on the tangent at D to the circle through A, B, C and D . PT cuts CA and CD at E and F respectively. $TF = TD$.

**Copy this diagram into your writing booklet.**

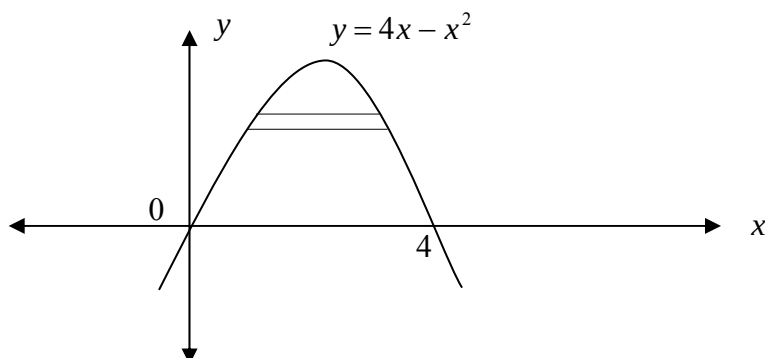
- (i) Show that $AEFD$ is a cyclic quadrilateral. 2
- (ii) Show that $PBEA$ is a cyclic quadrilateral. 3

Question 5 (15 Marks) Use a **SEPARATE** writing booklet.**Marks**

- (a) Find the general solution for the equation $\cos 3x = -\sin 2x$

3

- (b)

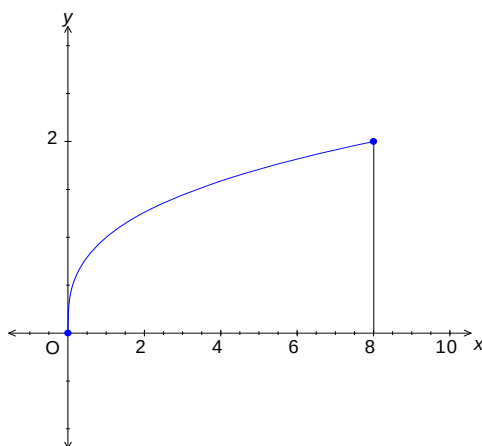


The area bounded by the curve $y = 4x - x^2$ and the x -axis is rotated about the y -axis.

- (i) Use strips perpendicular to the axis of rotation and show the x -coordinates of the end-points of these strips are $2 - \sqrt{4 - y}$ and $2 + \sqrt{4 - y}$.
- (ii) Find the maximum value y .
- (iii) Hence find the volume of the solid of revolution, in terms of π .

2**1****5**

- (c) The sketch below shows the region enclosed by the curve $y = x^{\frac{1}{3}}$, the x axis and the ordinate $x = 8$.



Find the volume generated when this region is rotated about the line $x = 8$, using the method of cylindrical shells.

4

Question 6 (15 Marks) Use a SEPARATE writing booklet.**Marks**

(a) Given that $z^n - \frac{1}{z^n} = 2i \sin n\theta$ and $\left(z - \frac{1}{z}\right)^5 = z^5 - 5z^3 + 10z - \frac{10}{z} + \frac{5}{z^3} - \frac{1}{z^5}$

(i) prove that: $\sin^5 \theta = \frac{1}{16} [\sin 5\theta - 5\sin 3\theta + 10\sin \theta]$ **3**

(ii) find the general solutions of the equation $16\sin^5 \theta = \sin 5\theta$. **4**

(b) A particle, of mass m , is projected vertically upwards in a resisted medium where the resistance is proportional to its velocity and mk is the constant of variation. The velocity of projection is given by $u \text{ ms}^{-1}$.

(i) Show that after a time t seconds, the height above the ground is:

$$x_1 = \frac{g + ku}{k^2} (1 - e^{-kt}) - \frac{gt}{k}. \quad \mathbf{5}$$

(ii) At the same time another particle is dropped from a height h metres vertically above the first particle. Given that at the time t seconds, its distance from the ground is:

$$x_2 = h + \frac{g}{k^2} (1 - e^{-kt}) - \frac{gt}{k},$$

show that the two particles will meet at a time T where

$$T = \frac{1}{k} \log \left(\frac{u}{u - kh} \right). \quad \mathbf{3}$$

Question 7 (15 Marks) Use a SEPARATE writing booklet.**Marks**

(a) i) Find the greatest and least values of e^{x^2-x} in the domain $0 \leq x \leq 2$. 2

ii) Hence show that $2e^{-\frac{1}{4}} < \int_0^2 e^{x^2-x} dx < 2e^2$ 1

(b) (i) Using the substitution $u = a - x$, prove that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$. 2

(ii) Hence show that $\int_0^\pi x \cos^2 x dx = \frac{\pi^2}{4}$. 3

(c) Given that $\cos(n\theta) + i \sin(n\theta) = (\cos \theta + i \sin \theta)^n$ where n is a positive integer,

(i) prove that

$$\cos(n\theta) = \cos^n \theta - \binom{n}{2} \cos^{n-2} \theta \sin^2 \theta + \binom{n}{4} \cos^{n-4} \theta \sin^4 \theta - \dots \text{and}$$

$$\sin(n\theta) = \binom{n}{1} \cos^{n-1} \theta \sin \theta - \binom{n}{3} \cos^{n-3} \theta \sin^3 \theta + \dots$$
 4

(ii) hence deduce that

$$\tan(n\theta) = \frac{\binom{n}{1} \tan \theta - \binom{n}{3} \tan^3 \theta + \dots}{1 - \binom{n}{2} \tan^2 \theta + \binom{n}{4} \tan^4 \theta - \dots}$$
 3

Question 8 (15 Marks) Use a SEPARATE writing booklet.

Marks

- (a) Use the compound angle formulae for $\cos(x+y)$ and $\cos(x-y)$ to prove the result

$$\cos S - \cos T = -2 \sin\left(\frac{S+T}{2}\right) \sin\left(\frac{S-T}{2}\right). \quad 2$$

(b) For $n = 0, 1, 2, 3, \dots$, define $I_n = \int_0^{\frac{\pi}{4}} \frac{1 - \cos 2nx}{\sin 2x} dx$.

- (i) Evaluate I_1 2

- (ii) Using the result proven in part (a), show that for $r \geq 1$:

$$I_{2r+1} - I_{2r-1} = \frac{1 - (-1)^r}{2r}. \quad 3$$

- (iii) Hence evaluate I_9 3

- (c) $P(a \cos \theta, b \sin \theta)$ and $Q(a \cos \phi, b \sin \phi)$ are points on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ whose eccentricity is e .

- (i) Find the equation of the chord PQ . 2

- (ii) If PQ is a focal chord of this ellipse show that $e = \frac{\sin(\phi - \theta)}{\sin \phi - \sin \theta}$. 3

END OF PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Solutions THGS Ext 2 Trial 2009

$$Q1a \int x^2 \sin(x^3) dx = -\frac{1}{3} \cos(x^3) + C$$

$$Q1b \int_0^1 \tan^{-1} x \, dx = \int_0^1 \tan^{-1} x \frac{d}{dx}(x) dx = \left[x \tan^{-1} x \right]_0^1 - \int_0^1 \frac{x}{1+x^2} dx$$

$$= \frac{\pi}{4} - 0 - \left[\frac{1}{2} \ln(1+x^2) \right]_0^1 = \frac{\pi}{4} - \frac{1}{2} \ln 2$$

$$Q1c \, i) \frac{x}{(x-1)(x+4)} \equiv \frac{a}{x-1} + \frac{b}{x+4}$$

$$\Rightarrow x \equiv a(x+4) + b(x-1)$$

$$\text{let } x=1 \Rightarrow a = \frac{1}{5}$$

$$\text{let } x=-4 \Rightarrow b = \frac{4}{5}$$

$$Q1cii) \int \frac{x}{(x-1)(x+4)} dx = \frac{1}{5} \int \frac{1}{x-1} dx + \frac{4}{5} \int \frac{1}{x+4} dx$$

$$= \frac{1}{5} \ln|x-1| + \frac{4}{5} \ln|x+4| + C$$

$$Q1d) \int \frac{x+4}{x^2-4x+13} dx = \frac{1}{2} \int \frac{2x-4}{x^2-4x+13} dx + 6 \int \frac{1}{(x-2)^2+9} dx$$

$$= \frac{1}{2} \ln|x^2-4x+13| + 2 \tan^{-1}\left(\frac{x-2}{3}\right) + C$$

$$Q1e) \text{ If } t = \tan \frac{x}{2} \text{ then } x = 2 \tan^{-1} t \text{ since } 0 \leq x \leq \frac{\pi}{2} \Rightarrow dx = \frac{2}{1+t^2} dt$$

$$\cos x = \frac{1-t^2}{1+t^2} \text{ and } x=0 \Rightarrow t=0, x=\frac{\pi}{2} \Rightarrow t=1$$

$$\therefore \int_0^{\frac{\pi}{2}} \frac{dx}{1+\cos x} = \int_0^1 \frac{\frac{2}{1+t^2} dt}{1+\frac{1-t^2}{1+t^2}} = \int_0^1 \frac{2dt}{1+t^2+1-t^2} = \int_0^1 dt = [t]_0^1 = 1$$

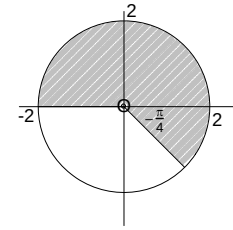
$$Q2a) w=1+i \quad z=1-i\sqrt{3}$$

$$i) w\bar{z} = (1+i)(1+i\sqrt{3}) = 1 - \sqrt{3} + i(1+\sqrt{3})$$

$$ii) \frac{1}{w} = \frac{1}{1+i} \times \frac{1-i}{1-i} = \frac{1-i}{2}$$

$$iii) \frac{i(\Re(z)-z)}{\Im(z)} = \frac{i(1-1+i\sqrt{3})}{-\sqrt{3}} = 1$$

$$Q2b \quad |z| \leq 2 \quad \pi \geq \arg z \geq -\frac{\pi}{4}$$



$$Q2c) x^2 - 4x + (1-4i) = 0$$

$$x = \frac{4 \pm \sqrt{16-4(1-4i)}}{2} = \frac{4 \pm \sqrt{12+16i}}{2} = 2 \pm \sqrt{3+4i}$$

$$\text{let } \sqrt{3+4i} = a+ib \text{ where } a, b \text{ are real} \quad \text{---(1)}$$

$$\therefore 3+4i = a^2 - b^2 + 2abi \Rightarrow a^2 - b^2 = 3 \quad \text{---(2)}, \left(2ab = 4 \Rightarrow b = \frac{2}{a} \quad \text{---(4)} \right)$$

$$\text{taking modulus of both sides of (1)} \Rightarrow 5 = a^2 + b^2 \quad \text{---(3)}$$

$$\text{Add (2) and (3)} \quad 2a^2 = 8 \Rightarrow a = \pm 2 \Rightarrow b = \pm 1$$

$$\left(\begin{array}{l} \text{OR using (2) and (4)} \\ a^2 - \left(\frac{2}{a}\right)^2 = 3 \Rightarrow (a^2)^2 - 3(a^2) - 4 = 0 \\ (a^2 - 4)(a^2 + 1) = 0 \Rightarrow a^2 = 4, \text{ reject } a^2 = -1 \text{ since } a \text{ is real} \\ \therefore a = \pm 2 \Rightarrow b = \pm 1 \end{array} \right)$$

$$\therefore x = 2 + (2+i) \text{ or } 2 - (2+i) \Rightarrow 4+i, \text{ or } -i$$

$$Q2di) \quad z = 1+2i+3t - (4t)i = 1+3t + i(2-4t)$$

$$\text{thus } x = 1+3t \quad \text{---(1)} \quad y = 2-4t \quad \text{---(2)}$$

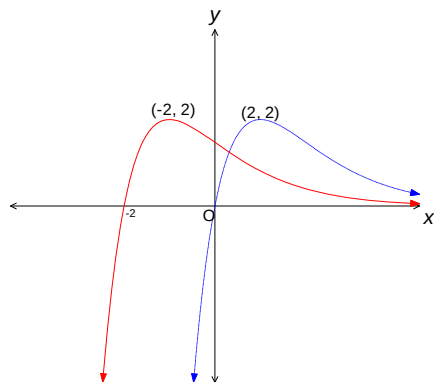
$$4 \times (1) + 3 \times (2) \Rightarrow 4x + 3y = 10$$

$$Q2dii) \quad |z| = \text{distance from origin}$$

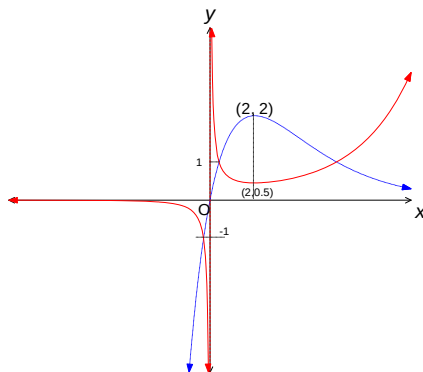
$$\therefore \min |z| = \text{perpendicular distance from origin}$$

$$= \frac{|4(0) + 3(0) - 10|}{\sqrt{4^2 + 3^2}} = 2$$

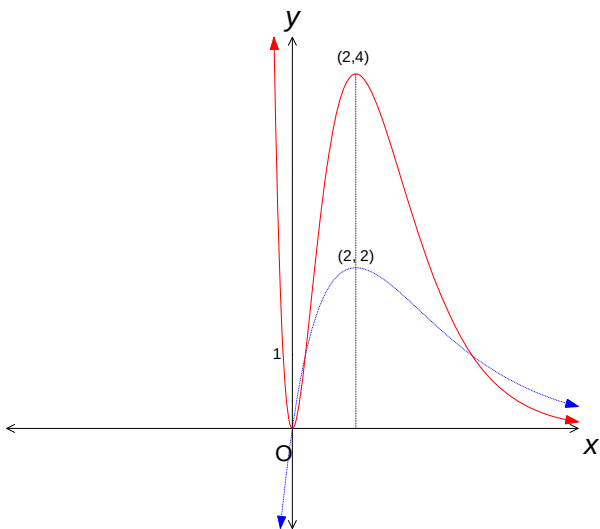
Q3ai)



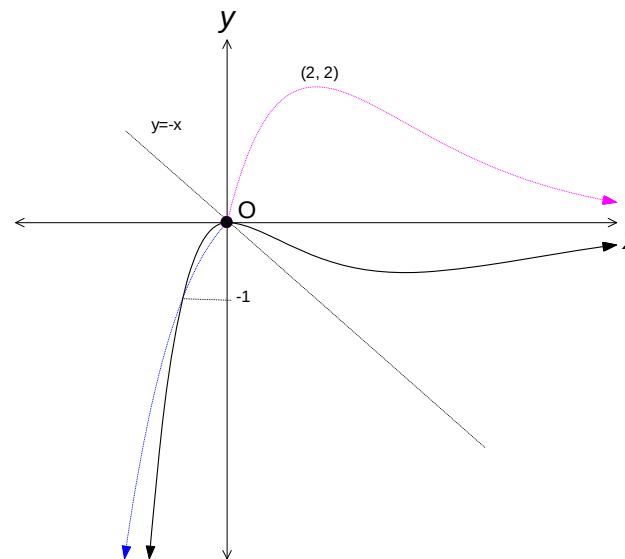
Q3aii)



Q3aiii)



Q3aiv)



$$Q3bi) \quad P(x) = x^5 - 3x^4 + 4x^3 - 4x + 3x - 1$$

$$P(1) = 1 - 3 + 4 - 4 + 3 - 1 = 0$$

$$P'(x) = 5x^4 - 12x^3 + 12x^2 - 8x + 3$$

$$P'(1) = 5 - 12 + 12 - 8 + 3 = 0$$

$$P''(x) = 20x^3 - 36x^2 + 24x - 8$$

$$P''(1) = 20 - 36 + 24 - 8 = 0$$

$\therefore P(1) = P'(1) = P''(1) = 0 \quad \therefore x = 1$ is a triple root

$$Q3bii) \quad P(i) = i^5 - 3i^4 + 4i^3 - 4i^2 + 3i - 1$$

$$= i - 3 - 4i + 4 + 3i - 1 = 0 \quad \therefore x = i$$
 is a root

Q3biii) If $x = i$ is a root so is $x = -i$

$\therefore$ roots are 1, 1, 1, i , $-i$

Q3c) let $y = \frac{1}{\sqrt{x}} \Rightarrow x = \frac{1}{y^2}$

Substitute in $x^3 - 2x^2 - 5x - 1 = 0$

$$\left(\frac{1}{y^2}\right)^3 - 2\left(\frac{1}{y^2}\right)^2 - 5\left(\frac{1}{y^2}\right) - 1 = 0$$

$$\frac{1}{y^6} - \frac{2}{y^4} - \frac{5}{y^2} - 1 = 0$$

$$\Rightarrow 1 - 2y^2 - 5y^4 - y^6 = 0$$

$$\therefore y^6 + 5y^4 + 2y^2 - 1 = 0 \text{ has roots } \frac{1}{\sqrt{\alpha}}, \frac{1}{\sqrt{\beta}}, \frac{1}{\sqrt{\gamma}}$$

Q4a) $\frac{x^2}{5-k} + \frac{y^2}{k-3} = 1$

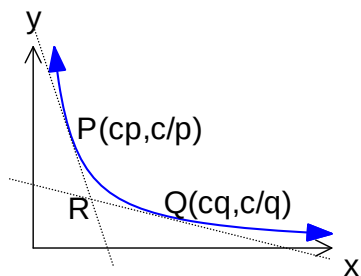
i) Circle if $5-k = k-3 \Rightarrow k = 4$

ii) Hyperbola if $5-k < 0$ and $k-3 > 0$

OR $5-k < 0$ and $k-3 > 0$

ie $(5-k)(k-3) < 0 \Rightarrow k > 5$ or $k < 3$

Q4bi)



$$xy = c^2 \Rightarrow y = \frac{c^2}{x} \therefore y' = -\frac{c^2}{x^2} = -\frac{1}{t^2} \text{ at } x = ct$$

$$\therefore \text{tangent is } y - \frac{c}{t} = \frac{-1}{t^2}(x - ct) \Rightarrow x + t^2y - 2ct = 0$$

Q4bii) Tangent at P $x + p^2y - 2cp = 0$ --- (1)

Tangent at Q $x + q^2y - 2cq = 0$ --- (2)

$$(1) - (2) \quad (p^2 - q^2)y = 2c(p - q)$$

$$y = \frac{2c}{p+q} \text{ as } p \neq q$$

$$\text{substitute in (1)} \quad x = 2cp - p^2 \left(\frac{2c}{p+q} \right) = \frac{2cp^2 + 2cpq - 2cp^2}{p+q}$$

$$\Rightarrow x = \frac{2cpq}{p+q}$$

$$\therefore R \text{ is } \left(\frac{2cpq}{p+q}, \frac{2c}{p+q} \right)$$

Q4biii) $\therefore x = \frac{2cpq}{p+q}$ --- (1)

$$y = \frac{2c}{p+q} \text{ --- (2)}$$

$$(1) \div (2) \quad \frac{x}{y} = pq \text{ and from (2) } p+q = \frac{2c}{y}$$

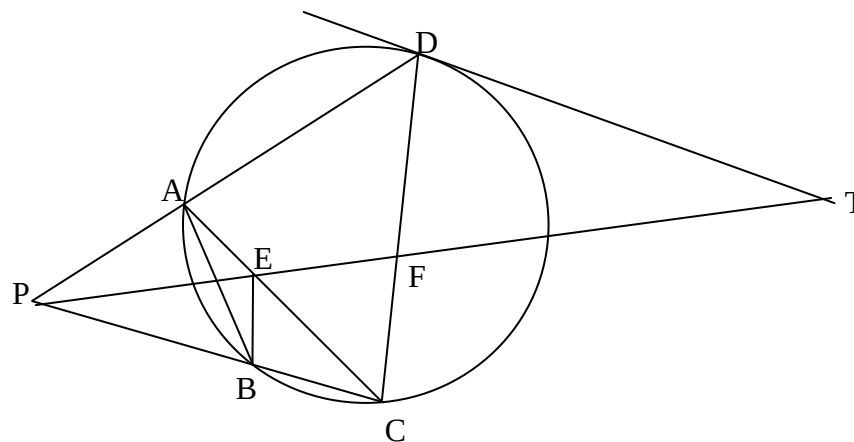
$$\text{Now } p^2 + q^2 = 2 \text{ given } \therefore (p+q)^2 - 2pq = 2$$

$$\left(\frac{2c}{y} \right)^2 - 2 \left(\frac{x}{y} \right) = 2 \Rightarrow 4c^2 - 2xy = 2y^2$$

$$\therefore xy + y^2 = 2c^2$$

OR substitute $R \left(\frac{2cpq}{p+q}, \frac{2c}{p+q} \right)$ in $xy + y^2 = 2c^2$ to show true.

Q4c)



i) $TD = TF$ given

$\therefore \angle TFD = \angle TDF$ base angles of isosceles triangle are equal

$\angle TDF = \angle CAD$ angle between tan gent and chord at point of contact equals angle in the alternate segment

$\therefore \angle TFD = \angle CAD$

$\therefore AEFD$ is cyclic quad since exterior angle equals interior opposite angle

ii) $\angle PEA = \angle ADF$ exterior angle of cyclic quad $AEFD$ equals interior opposite angle

$\angle PBA = \angle ADF$ exterior angle of cyclic quad $ABCD$ equals interior opposite angle

$\therefore \angle PEA = \angle PBA$

$\therefore PBEA$ is cyclic since these two angles stand on the interval AP and are on the same side of the interval

Q5a) $\cos 3x = -\sin 2x = \sin(-2x)$ since odd function

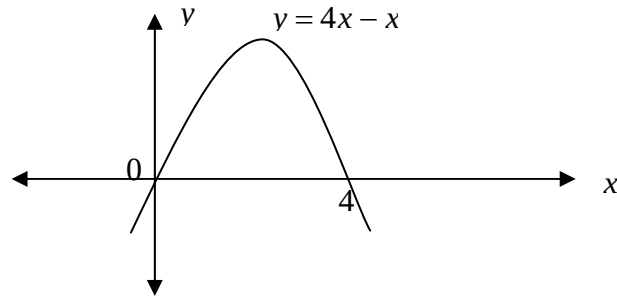
$$\therefore \cos 3x = \cos\left(\frac{\pi}{2} + 2x\right)$$

$$\Rightarrow 3x = 2n\pi \pm \left(\frac{\pi}{2} + 2x\right)$$

$$\therefore x = \frac{\pi}{2}(4n+1) \text{ or } 5x = \frac{\pi}{2}(4n-1)$$

$$\Rightarrow x = \frac{\pi(4n+1)}{2} \text{ or } x = \frac{\pi(4n-1)}{10}$$

Q5b

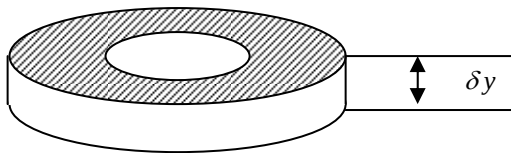


i) $x^2 - 4x + 4 = 4 - y$

$$x - 2 = \pm\sqrt{4 - y} \Rightarrow x = 2 + \sqrt{4 - y}, 2 - \sqrt{4 - y}$$

ii) $y_{\max} = f(2) = 8 - 4 = 4$

iii) Rotating the slice indicated creates a disc



$$\text{Volume of disc} = \delta V = \pi(x_2^2 - x_1^2)\delta y$$

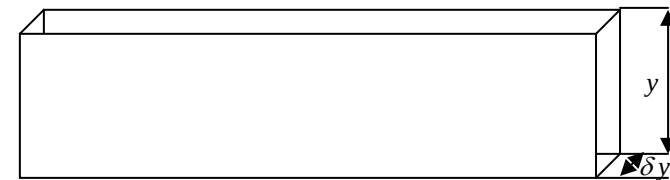
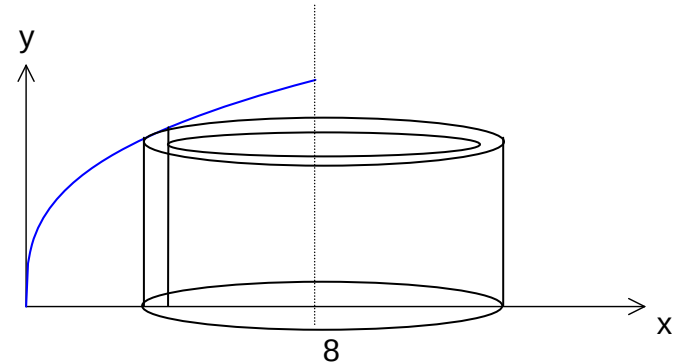
$$V = \lim_{\delta y \rightarrow 0} \sum_{y=0}^{y=4} \pi(x_2^2 - x_1^2)\delta y$$

$$= \pi \int_0^4 (x_2^2 - x_1^2) dy = \pi \int_0^4 (x_2 - x_1)(x_2 + x_1) dy$$

$$= \pi \int_0^4 2\sqrt{4-y} \times 4 dy = 8\pi \int_0^4 (4-y)^{\frac{1}{2}} dy$$

$$= 8\pi \times \frac{-2}{3} \left[(4-y)^{\frac{3}{2}} \right]_0^4 = 8\pi \times \frac{-2}{3} [0 - 8] = \frac{128\pi}{3} \text{ unit}^3$$

Q5c



$$y = x^{\frac{10}{3}}$$

$$2\pi(8-x)$$

$$\text{Volume of shell} = \delta V \approx 2\pi(8-x)y\delta x = 2\pi(8-x)x^{\frac{1}{3}}\delta x$$

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^{x=8} \delta V = 2\pi \int_0^8 \left(8x^{\frac{1}{3}} - x^{\frac{4}{3}} \right) dx = 2\pi \left[8 \times \frac{3}{4} x^{\frac{4}{3}} - \frac{3}{7} x^{\frac{7}{3}} \right]_0^8$$

$$= 2\pi \left(6 \times 16 - \frac{3}{7} \times 128 \right) = \frac{576\pi}{7} \text{ unit}^3$$

$$\begin{aligned} \text{Q6ai)} \text{ Given } \left(z - \frac{1}{z}\right)^5 &= z^5 - 5z^3 + 10z - \frac{10}{z} + \frac{5}{z^3} - \frac{1}{z^5} \\ &= z^5 - \frac{1}{z^5} - 5\left(z^3 - \frac{1}{z^3}\right) + 10\left(z + \frac{1}{z}\right) \end{aligned}$$

$$\therefore (2i \sin \theta)^5 = 2i(\sin 5\theta - 5\sin 3\theta + 10\sin \theta)$$

$$\therefore \sin^5 \theta = \frac{1}{16}(\sin 5\theta - 5\sin 3\theta + 10\sin \theta)$$

$$\text{Q6aii)} \text{ from i) } 16\sin^5 \theta = \sin 5\theta - 5\sin 3\theta + 10\sin \theta$$

$$\text{if } 16\sin^5 \theta = \sin 5\theta \text{ then } -5\sin 3\theta + 10\sin \theta = 0$$

$$\begin{aligned} \therefore 2\sin \theta &= \sin 3\theta = \sin(2\theta + \theta) = \sin 2\theta \cos \theta + \cos 2\theta \sin \theta \\ &= 2\sin \theta \cos^2 \theta + (1 - 2\sin^2 \theta)\sin \theta = 3\sin \theta - 4\sin^3 \theta \end{aligned}$$

$$\therefore 4\sin^3 \theta - \sin \theta = 0 \Rightarrow \sin \theta(4\sin^2 \theta - 1) = 0$$

$$\sin \theta = 0, \quad \pm \frac{1}{2} \Rightarrow \theta = n\pi, \quad n\pi \pm \frac{\pi}{6} \text{ for integer } n$$

$$\text{Q6b)} \quad R = -mg - mkv \quad \therefore \ddot{x} = -g - kv \quad \uparrow +ve \quad \downarrow mg \downarrow mkv$$

$$\frac{dv}{dt} = \frac{-1}{g + kv}$$

$$\therefore t = \int_u^v \frac{-1}{g + kv} dv \Rightarrow t = \left[\frac{-1}{k} \ln |g + kv| \right]_u^v$$

$$t = \frac{-1}{k} \ln \left| \frac{g + kV}{g + ku} \right| \Rightarrow -kt = \ln \left| \frac{g + kV}{g + ku} \right|$$

$$\frac{g + kV}{g + ku} = e^{-kt} \Rightarrow g + kV = (g + ku)e^{-kt} \therefore V = \left(\frac{g + ku}{k} \right) e^{-kt} - \frac{g}{k}$$

$$\therefore x = \int_0^t \left(\left(\frac{g + ku}{k} \right) e^{-kt} - \frac{g}{k} \right) dt = \left[-\left(\frac{g + ku}{k^2} \right) e^{-kt} - \frac{gt}{k} \right]_0^t$$

$$\therefore x = \left(\frac{g + ku}{k^2} \right) (1 - e^{-kt}) - \frac{gt}{k}$$

$$\text{Q6bii)} \quad t = T \text{ and } x_1 = x_2$$

$$\therefore \frac{g + ku}{k^2} (1 - e^{-kT}) - \frac{gT}{k} = h + \frac{g}{k^2} (1 - e^{-kT}) - \frac{gT}{k}$$

$$(1 - e^{-kT}) \left[\frac{g + ku}{k^2} - \frac{g}{k^2} \right] = h$$

$$1 - e^{-kT} = \frac{hk}{u} \Rightarrow e^{-kT} = \frac{u - hk}{u}$$

$$\therefore -kT = \ln \left(\frac{u - hk}{u} \right)$$

$$T = \frac{1}{k} \ln \left(\frac{u}{u - hk} \right)$$

$$\text{Q7ai)} \quad \text{Consider } y = e^{x^2 - x}$$

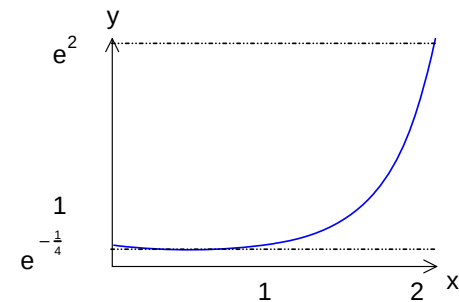
$$y' = (2x - 1)e^{x^2 - x} \Rightarrow y' = 0 \text{ when } x = \frac{1}{2}$$

$$\begin{array}{ccccc} x & 0 & \frac{1}{2} & 1 & \\ y' & -ve & 0 & +ve & \end{array} \therefore \text{Minimum at } \left(\frac{1}{2}, e^{-\frac{1}{4}} \right)$$

$$\text{also } f(0) = e^0 = 1 \quad f(2) = e^2 \quad \therefore \text{Maximum at } (2, e^2)$$

$$\text{areaOABC} < \int_0^2 e^{x^2 - x} dx < \text{areaOADE}$$

$$\therefore 2e^{-\frac{1}{4}} < \int_0^2 e^{x^2 - x} dx < 2e^2$$



$$Q7bi) \quad \text{let } u = a - x \Rightarrow du = -dx, \quad x = 0 \Rightarrow u = a, \quad x = a \Rightarrow u = 0$$

$$\therefore \int_0^a f(a-x) dx = \int_a^0 f(u) \times -du = \int_0^a f(u) du = \int_0^a f(x) dx$$

$$ii) \quad \int_0^\pi x \cos^2 x dx = \int_0^\pi (\pi - x) \cos^2 (\pi - x) dx = \int_0^\pi (\pi - x) \cos^2 x dx$$

$$\therefore 2 \int_0^\pi x \cos^2 x dx = \pi \int_0^\pi \cos^2 x dx = \frac{\pi}{2} \int_0^\pi (\cos 2x + 1) dx$$

$$= \frac{\pi}{2} \left[\frac{1}{2} \sin 2x + x \right]_0^\pi = \frac{\pi}{2} (\pi - 0)$$

$$\therefore \int_0^\pi x \cos^2 x dx = \frac{\pi^2}{4}$$

$$Q7c) \quad \cos n\theta + i \sin n\theta = (\cos \theta + i \sin \theta)^n$$

$$= \cos^n \theta + \binom{n}{1} \cos^{n-1} \theta (i \sin \theta) + \binom{n}{2} \cos^{n-2} \theta (i \sin \theta)^2 + \binom{n}{3} \cos^{n-3} \theta (i \sin \theta)^3 + \dots$$

$$= \cos^n \theta - \binom{n}{2} \cos^{n-2} \theta \sin^2 \theta + \binom{n}{4} \cos^{n-4} \theta \sin^4 \theta - \dots$$

$$+ i \left(\binom{n}{1} \cos^{n-1} \theta \sin \theta - \binom{n}{3} \cos^{n-3} \theta \sin^3 \theta + \dots \right)$$

$$\text{using } i^2 = -1, i^3 = -i, i^4 = 1, \dots$$

Equating real and imaginary parts

$$\cos n\theta = \cos^n \theta - \binom{n}{2} \cos^{n-2} \theta \sin^2 \theta + \binom{n}{4} \cos^{n-4} \theta \sin^4 \theta - \dots \quad \text{--- (1)}$$

$$\sin n\theta = \binom{n}{1} \cos^{n-1} \theta \sin \theta - \binom{n}{3} \cos^{n-3} \theta \sin^3 \theta + \dots \quad \text{--- (2)}$$

$$ii) (2) \div (1) \quad \frac{\sin n\theta}{\cos n\theta} = \tan n\theta = \frac{\binom{n}{1} \cos^{n-1} \theta \sin \theta - \binom{n}{3} \cos^{n-3} \theta \sin^3 \theta + \dots}{\cos^n \theta - \binom{n}{2} \cos^{n-2} \theta \sin^2 \theta + \binom{n}{4} \cos^{n-4} \theta \sin^4 \theta - \dots}$$

divide top and bottom by $\cos^n \theta$

$$\tan n\theta = \frac{\binom{n}{1} \tan \theta - \binom{n}{3} \tan^3 \theta + \dots}{1 - \binom{n}{2} \tan^2 \theta + \binom{n}{4} \tan^4 \theta - \dots}$$

$$Q8a) \quad \cos(x+y) = \cos x \cos y - \sin x \sin y \quad \text{--- (1)}$$

$$\cos(x-y) = \cos x \cos y + \sin x \sin y \quad \text{--- (2)}$$

$$(1) - (2) \Rightarrow \cos(x+y) - \cos(x-y) = -2 \sin x \sin y$$

$$\text{let } x+y = S, \quad x-y = T \Rightarrow x = \frac{S+T}{2}, \quad y = \frac{S-T}{2}$$

$$\therefore \cos S - \cos T = -2 \sin \frac{S+T}{2} \sin \frac{S-T}{2}$$

$$Q8b) \quad I_n = \int_0^{\frac{\pi}{4}} \frac{1 - \cos 2nx}{\sin 2x} dx$$

$$i) \quad I_1 = \int_0^{\frac{\pi}{4}} \frac{1 - \cos 2x}{\sin 2x} dx = \int_0^{\frac{\pi}{4}} \frac{2 \sin^2 x}{2 \sin x \cos x} dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{\sin x}{\cos x} dx = -[\ln |\cos x|]_0^{\frac{\pi}{4}} = -\ln \frac{1}{\sqrt{2}} = \frac{1}{2} \ln 2$$

$$Q8bii) \quad I_{2r+1} - I_{2r-1} = \int_0^{\frac{\pi}{4}} \left(\frac{1 - \cos(4xr + 2x)}{\sin 2x} - \frac{1 - \cos(4xr - 2x)}{\sin 2x} \right) dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{-2 \sin 4xr \cdot \sin(-2x)}{\sin 2x} dx \quad \text{from (a)}$$

$$= 2 \int_0^{\frac{\pi}{4}} \sin 4xr dx = \frac{-1}{2r} [\cos 4xr]_0^{\frac{\pi}{4}} = \frac{-1}{2r} [\cos r\pi - 1]$$

$$= \frac{-1}{2r} [(-1)^r - 1] = \frac{1 - (-1)^r}{2r}$$

$$Q8biii) \quad I_9 = I_{2 \times 4 + 1} \therefore r = 4 \Rightarrow I_9 - I_7 = 0 \quad \text{--- (1)}$$

$$r = 3 \Rightarrow I_7 - I_5 = \frac{1+1}{6} = \frac{1}{3} \quad \text{--- (2)}$$

$$r = 2 \Rightarrow I_5 - I_3 = 0 \quad \text{--- (3)}$$

$$r = 1 \Rightarrow I_3 - I_1 = \frac{1+1}{2} = 1 \quad \text{--- (4)}$$

$$(1) + (2) + (3) + (4) \Rightarrow I_9 - I_1 = \frac{4}{3}$$

$$I_1 = \frac{1}{2} \ln 2 \quad \text{from i)}$$

$$\therefore I_9 = \frac{4}{3} + \frac{1}{2} \ln 2$$

$$Q8ci) \quad P(a \cos \theta, b \sin \theta) \quad Q(a \cos \phi, b \sin \phi)$$

$$m_{PQ} = \frac{b(\sin \phi - \sin \theta)}{a(\cos \phi - \cos \theta)}$$

$$\therefore \text{equation of chord } PQ \text{ is } y - b \sin \theta = \frac{b(\sin \phi - \sin \theta)}{a(\cos \phi - \cos \theta)}(x - a \cos \theta)$$

$$\Rightarrow ay(\cos \phi - \cos \theta) - ab \sin \theta(\cos \phi - \cos \theta) = bx(\sin \phi - \sin \theta) - ab \cos \theta(\sin \phi - \sin \theta)$$

$$ii) \quad \text{Focal chord through } (ae, 0)$$

$$\therefore -ab \sin \theta(\cos \phi - \cos \theta) = bae(\sin \phi - \sin \theta) - ab \cos \theta(\sin \phi - \sin \theta)$$

$$e(\sin \phi - \sin \theta) = \cos \theta \sin \phi - \cos \theta \sin \theta - \sin \theta \cos \phi + \sin \theta \cos \theta$$

$$= \sin(\phi - \theta)$$

$$\Rightarrow e = \frac{\sin(\phi - \theta)}{(\sin \phi - \sin \theta)}$$